

Slide 1

Gen AI Enabled Green Coding Solution

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Slide 2

Idea – Green IT Across Tech Value Chain

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Ambition : Most of the Customers have Net Zero Target by ~2040 to support the Sustainability.

■■ TCS has Net Zero target by 2030

There is increased demand for bringing in Green IT principles. There are options of bringing sustain

Cloudification is going to directly influence sustainability goals

Green IT incl Green Coding, Optimized developments and reducing vulnerabilities

The Vision is to create a Gen AI accelerator which can help finding vulnerabilities related to Green

Following are few of features which can be checked as part of accelerator:

Code Performance issues or prospect issues in the legacy code

Lean coding Principles

Virtual machines/Containerization

Bad design/Technical Debt

Use energy-efficient libraries and frameworks

Efficient algorithms and data structures

Reduce power consumption during the debugging process

Slide 3

Solution Approach – Option1

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Building & Training Generative AI Model

(LLM/Custom Model)

PEGA

Python

Mendix

JAVA

REST API

Custom

API

Revised Code

Code Snippet

LLM Training for Energy Efficiency : Data Input and Preparation, Data Ingestion, Data Cleaning, Norm

Azure Open AI

Amazon Bedrock

GCP

TCS Fine Tuned LLM ?

Agentic

Approach

Create an API based tool which can work as Green Coding Reviewer for multiple applications/technolog

Apply Green Coding

Plug-Ins

Green Coding Defects

Analysis

Reasoning

Slide 4

Improve Green Coding Coverage

When run on an application, the tool highlights for review the Green Coding design decisions Powered Process Flow

Take a Back-Up and Database Point Restore of the environment before installation

Login to Pega Designer Studio & Import utility

Run the Utility which Scan the database and Invoke Pega OOTB LLM (Pre-trained for Green Coding) and

Observations are listed in Report structure that which can be saved for offline analysis

Agentic Approach provides suggestions for Green Code and once accepted will overwrite the previously Approach

Option2: Installable Utility for each Low Code-No Code Platform

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Sample Flow for Pega

Invoke Pega Gen AI

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Invoke LLM (Options:)

Amazon Bedrock

GCP – Vertex AI

Grok.AI

Llama

Pega provides integrations with Azure OpenAI Service, Vertex AI, and Amazon AWS Bedrock to access va Improvement

Suggestions

Agentic

Approach

SelectApp

Select

Rulesets

Apply Green Coding

Plug-Ins

Process

Results

View

Results

Slide 5

Growth by 2030

Sept 2023

Euro 4069 million

India ■ Market Reach

15%

CAGR

Overall

23%

Market Segment ■ Growth EU

MM

Revenue Increase

FUTURE

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Effort required to build MVP solution ~3 months and can be taken to Marketplace

Slide 6

The TCS commitment

In it for good.

We take a long-term view, building relationships that endure, leading to mutual growth and sustainab

Bring everything.

When you are out to change your world, you need a partner who can combine every resource in their ar

Know-how.

With TCS, you experience certainty. We are a knowledge network of the smartest minds and practitione

Master the journey.

Along with certainty, you also need partners who can deal with

uncertainty. We provide the agility and continuity you need to navigate perpetual change.

Digital transformation requires a holistic approach with intelligent solutions